Khaula Molapo

20240001

Week: 6

Software Project Management

1.

In a mobile app development project, the project manager must organize the work clearly to ensure the project is completed on time. A Work Breakdown Structure (WBS) is created to divide the project into smaller and manageable tasks. The main project is divided into major phases such as planning, design, development, testing, and deployment.

Each phase is further divided into smaller tasks. For example, the design phase includes user interface design and database design. The development phase includes frontend coding, backend coding, and integration. After creating the WBS, the project manager develops a schedule by assigning start dates, end dates, and task durations.

Dependencies between tasks are identified. For example, testing cannot begin before development is completed. By organizing tasks using WBS and scheduling them properly, the team can track progress, manage time effectively, and reduce delays. This approach improves project control and ensures that project objectives are achieved.

2.

The Critical Path Method (CPM) is a project management technique used to identify the longest sequence of dependent tasks in a project. This sequence is called the critical path. The critical path determines the shortest time required to complete the project.

If any task on the critical path is delayed, the entire project will be delayed. CPM helps project managers identify which tasks are most important and require close monitoring. It also helps in calculating task durations and understanding task dependencies.

By using CPM, managers can focus on high-risk tasks, allocate resources effectively, and reduce project delays. CPM improves planning accuracy and supports better decision-making in software project management.

3.

In GanttProject, I created a new project and entered the main tasks based on the WBS. I added tasks such as planning, design, development, testing, and deployment. For each task, I

assigned start dates, durations, and dependencies.

The Gantt chart automatically displayed the tasks in a timeline format. I linked dependent tasks to show the correct order of execution. This helped me see how tasks overlap and how long the entire project will take.

From this activity, I learned that Gantt charts provide a clear visual representation of project schedules. They help track progress and identify possible delays. Using GanttProject makes scheduling easier and improves time management in software projects.

4.

In Draw.io, the WBS flowchart begins with the main project at the top. Below it, major phases such as planning, design, development, testing, and deployment are displayed as separate branches.

Each major phase is divided into smaller tasks. For example, development can include frontend development and backend development. These tasks are shown as child elements connected to their parent phase.

The diagram clearly shows the hierarchical structure of the project. It helps the team understand how the project is organized and what tasks need to be completed. The WBS flowchart improves clarity, planning, and communication among team members.